

WEEKLY REPORT

Work done in last week:

- Built an NVIDIA NIM text embedder for semantic search.
- Set up Qdrant vector store with an LLM reranker for improved retrieval.
- Developed a hybrid search (semantic + keyword) using RRF for better ranking.
- Implemented input guardrails to detect prompt injection, validate topics, and use AI for uncertain cases.
- Implemented output guardrails to remove sensitive data before responses are sent.
- Added an agent handover protocol with audit logging for all transfers.

Plans for next week:

- Build the BaseAgent class and the Triage Agent that reads messages and decides which agent should handle them.
- Build the Technical Agent and Billing Agent with knowledge base search.
- Connect all agents together using the LangGraph orchestrator..

References:

1. NVIDIA NIM API Documentation. Retrieved from:
<https://build.nvidia.com/>
2. Qdrant Documentation. Retrieved from: <https://qdrant.tech/documentation/>
3. Reciprocal Rank Fusion — Cormack et al. Retrieved from:
<https://dl.acm.org/doi/10.1145/1571941.1572114>
4. OWASP LLM Top 10. Retrieved from:
<https://owasp.org/www-project-top-10-for-large-language-model-applications/>



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